



KOLEHIYO NG LUNGSOD NG DASMARIÑAS

**Development of Students Record Management System for Kolehiyo ng
Lungsod ng Dasmariñas - Registrar office**

AN UNDERGRADUATE CAPSTONE PROPOSAL

**Presented to the Faculty of the
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**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Systems**

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Chapter 2

REVIEW OF RELATED LITERATURE

In this chapter, the focus is on reviewing literature and studies related to the Development of Students Record Management System for Kolehiyo ng Lungsod ng Dasmariñas - Registrar office. It aims to provide a thorough understanding of the current research, technologies, and approaches that are critical to the design, functionality, and improvement of the proposed system. The discussion includes frameworks, methodologies, and insights from previous studies that serve as a foundation for developing an efficient and reliable records management system.

Foreign and Local Related Literature

1.1 Student Records Management

Record management plays a very important role in schools, especially in managing and organizing student information. With the rapid development of technology, many educational institutions are now adopting student record management systems to improve the way they store and handle student data. These systems allow schools to securely store and manage important information such as students' personal details, academic records, grades, enrollment data, and certificates of enrollment in an organized database.

The management of student records is an essential part of academic institutions because it supports many administrative decisions such as course offerings, faculty performance, and class scheduling. Proper record management



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ensures that records, whether in physical or digital form, are well maintained, easily retrieved, and continuously updated. According to the ISO 15489-1:2016 Records Management Standard, records are information assets that serve as evidence of organizational activities and are important for business operations. Poor record management can lead to data inaccuracies, security risks, and potential damage to an institution's reputation. Therefore, educational institutions must implement effective record management practices to ensure secure, accessible, and reliable information for both staff and students. (Reyes et al, 2023).

1.2 Records Management in Educational Institutions

Higher education institutions keep a variety of records that are crucial to institutional operations and decision-making, including financial data, staff and student information, and academic policies. However, a lot of processing time, labor, and storage space are needed for standard record-keeping procedures. Using the UTAUT and TOE models, the proposed system seeks to enhance efficiency and technology adoption in record management by managing both academic and non-academic activities, such as student records, attendance, inventory, library, and financial data, (K. Thilakavalli, A. Saraswathi, D. Rajagopal, 2023).

In higher education, digital record management boosts productivity, increases accuracy, and advances organizational objectives. The results demonstrated that the organization already has an efficient digital records management system that improves operational performance and staff engagement. To further enhance the



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institution's digital transformation, decision-making, and system efficiency, the researchers suggested creating a strategic plan (Wagan et al., 2025).

1.3 Computerized Student Information System

A Computerized Student Information System (SIS) is a digital system designed to manage and store student data electronically. These systems help educational institutions maintain accurate records and improve the efficiency of administrative processes. According to Das et al. (2023), student information systems are widely used in schools and universities to manage student data such as enrollment details, grades, attendance records, and academic history. The researchers explained that computerized systems allow institutions to access and update student information quickly, which helps reduce manual workload and improve data accuracy.

1.4 Database Management System

A Database Management System (DBMS) is software that allows users to store, organize, and manage large amounts of data efficiently. In modern information systems, databases play an important role in managing digital information. According to John et al. (2022), database management systems are essential components of student information systems because they allow institutions to store large volumes of student data in an organized and structured manner. The study explained that databases help ensure data consistency, reduce redundancy, and allow administrators to retrieve information quickly.



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Foreign and Local Related Studies

2.1 Students Record Management

Student records management refers to the process of collecting, organizing, storing, and maintaining student information within an educational institution. These records usually contain personal details, academic performance, enrollment history, and other important documents related to a student's education. Proper management of these records is important because it helps institutions maintain accurate information and ensures that records can be easily retrieved when needed. According to Riño and Daing (2022), student records management plays an important role in supporting academic administration because it allows institutions to maintain organized and reliable information about their students. The study explained that when records are not properly managed, institutions may experience problems such as missing documents, slow retrieval of information, and inaccurate data management.

In order to facilitate the digitalization of academic records, Falolo, Capillas, Vergarra, and Cerbito (2022) investigated the use of student registration and records management systems. The researchers employed a validated questionnaire to gather data from participants in a descriptive-evaluative study design. According to the results, students had a moderate amount of trouble with missing enrollment deadlines and admission requirements. To improve efficiency and record



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management in educational institutions, the study suggested making enrollment services more accessible and digitizing student data.

2.2 Records Management in Educational Institutions

Yidana and Akuna (2025) reviewed academic records management practices in higher education institutions across Africa using a scoping review of 65 studies from 15 countries. The study examined record management processes, the effectiveness of digital record systems, challenges in implementation, and measures for protecting the privacy and security of student records. The findings revealed that digital record systems improve efficiency, but challenges such as limited IT infrastructure, lack of digital skills, and weak policy implementation affect effective records management.

According to Harits Ar Rosyid (2024), The SIONLAP 2.0 system was developed to digitize laboratory management processes in higher education institutions by improving data management, workflow processing, and user access control. Although the system focuses on laboratory asset and inventory management, it demonstrates how information systems can improve efficiency and organization of institutional records. This idea is connected to the creation of a system for managing student records, which also seeks to digitize and arrange academic data in educational Institutions.

2.3 Computerized Students Information System



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Silwamba and Matela (2024) developed a digital student record system designed to improve the management of academic records in educational institutions. The study showed that computerized systems help institutions manage student information more efficiently compared to manual record-keeping methods.

2.4 Database Management System

Data management today is one of the most important processes in organizing and storing information such as personal or work-related data. In this study, keeping student records and personal information is important to ensure that they are stored in a secure database. These records should only be accessed by authorized personnel, such as staff in the registrar's office, to protect the confidentiality and security of student information. Educational institutions must implement proper access control to prevent unauthorized users from viewing or modifying sensitive data. MySQL is a database management system that supports corporate demands for data storage and management by providing flexibility, high performance, and dependable security, (Umami Kalsum Siregar et al, 2024).

2.5 Technical System Background

The development of the proposed system uses several technologies including PHP, MySQL, JavaScript, Bootstrap, and Visual Studio Code. According to Welling and Thomson (2023), PHP is a widely used server-side scripting language that allows developers to create dynamic web applications and connect them to databases such



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as MySQL. This technology is commonly used in web-based information systems due to its flexibility and compatibility with many platforms..

Related systems

IPAS University of Washington IT Services Team (2021) includes information and services that educate students about educational opportunities and resources. It consists of in order to improve retention and graduation rates, early warning systems are used to identify students who may face academic difficulties.

Christian Childcare Programmed Vocational Training Institute (2022) Ability to effectively manage data. The Student Record Management System aims to create a comprehensive database for student records, significantly reducing administrative workload, enhancing data accuracy, and streamlining report generation. By adopting modern tools and an organized approach, the system promises improved efficiency and user satisfaction.

Savitribai Phule Pune University, Pune, Maharashtra, India (2021) Collected feedback from students and academic staff shows that such a system can help to maintain an effective and efficient supervision relationship between all parties. Traditional assignment management is becoming obsolete due to its inconvenience, inefficiency, and low accuracy. Currently, web-based management systems have been widely implemented due to the development of Web technologies.



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This project has many features which are generally not available in normal record management systems like student login. It also has a whole system. It has functionality where students after logging into their id's can see detailed information about them by giving entry id and the corresponding password. This project develops to help the student as well as staff of the academic to maintain the records in the best way possible and also to reduce the manual. Dhenkanal Autonomous College (2021)

The University of Louisiana at Monroe (2021) is an online, open-source application for student achievement, degree auditing, and academic guidance. It offers tools for making appointments, monitoring academic achievement, and evaluating student risk factors, encouraging proactive approaches to intervention.

Project Synthesis

The gathered Review of Related Literature and Studies (RRLS) shows that many educational institutions are adopting computerized or digital record management systems to improve the storage, organization, and retrieval of student data. Similar to the proposed Development of a Web-Based Students Record Management System for the Kolehiyo ng Lungsod ng Dasmariñas (KLD) Registrar's Office, previous studies emphasize the importance of digitizing academic records to increase efficiency, accuracy, and security. Another similarity between the proposal and the reviewed literature is the goal of improving administrative processes in educational institutions. Many related systems aim to simplify record handling,



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reduce paperwork, and allow faster access to information. The proposed system also supports these objectives by providing features such as online document requests, grade posting by professors, automatic section creation, and centralized database storage. These functions align with the findings of several studies that digital systems improve efficiency and accessibility of student records in schools and universities.

However, there are also differences between the proposed system and the gathered RRLS. Some related systems focus on broader institutional functions such as laboratory management, academic advising, early warning systems for student performance..On another side, the proposed project focuses on KLD's Registrar's Office, particularly in managing student academic records and processing document requests such as Transcript of Records (TOR), Certificate of Enrollment (COE), and Certificate of Grades (COG). The system is also limited to authorized users within the institution and does not include other departments like accounting or admissions.

In summary, the gathered RRLS supports the importance of digital record management systems in educational institutions. While many studies highlight the general benefits of computerized student information systems, the proposed project differs by addressing the specific needs of the KLD Registrar's Office and providing a customized solution for managing student records, improving service efficiency, and ensuring secure and organized academic data management.



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